

YEAR 11 SEMESTER 1 PHYSICS DATA AND EQUATION SHEET

The following tables show the symbols of common quantities and physical constants used in equations. Vectors are shown by placing arrows over their symbols. If only the magnitude of a vector quantity is used, the arrow is not used.

Symbols of Common Quantities

heat Q	distance d	wavelength λ	charge q
specific heat capacity c	change in time Δt	frequency f	potential difference ΔV
temperature T	displacement $\vec{s}$	wave speed v	current I
temperature change ΔT	initial velocity $\vec{v}_o$	period T	resistance R
latent heat L	final velocity $\vec{v}$	integer m	resistivity ρ
mass m	acceleration $\vec{a}$		cross-sectional area A
work W	time t		length l
power P	drag force $\vec{F}_D$		
	drag coefficient C_D		
	density ρ		
	area A		

Physical Constants

Acceleration due to gravity on Earth's surface $g = 9.80 \text{ m s}^{-2}$

Speed of light in a vacuum $c = 3.00 \times 10^8 \text{ m s}^{-1}$

Charge of the electron $e = 1.60 \times 10^{-19} \text{ C}$

Equations

Heat

$$T(\text{K}) = T(^{\circ}\text{C}) + 273$$

$$Q = mc\Delta T$$

$$Q = mL$$

$$P = \frac{W}{\Delta t} = \Delta V \times I$$

Linear Motion

$$\text{average speed} = \frac{d}{\Delta t}$$

$$\vec{v}_{av} = \frac{\vec{s}}{\Delta t}$$

$$\vec{a}_{av} = \frac{\Delta \vec{v}}{\Delta t} = \frac{\vec{v} - \vec{v}_o}{\Delta t}$$

$$\vec{v} = \vec{v}_o + \vec{a}t$$

$$\vec{s} = \vec{v}_o t + \frac{1}{2} \vec{a}t^2$$

$$v^2 = v_o^2 + 2as$$

$$\vec{v}_{av} = \frac{\vec{v}_o + \vec{v}}{2} = \frac{\vec{s}}{t}$$

$$\vec{F}_D = \frac{1}{2} C_D \rho A v^2$$

Waves

$$f = \frac{1}{T}$$

$$v = f\lambda$$

Constructive interference:

$$\text{Path difference} = m\lambda$$

Destructive interference:

path difference

$$= \left(m + \frac{1}{2}\right)\lambda$$

Electricity

$$\Delta V = \frac{W}{q}$$

$$I = \frac{q}{\Delta t}$$

$$R = \frac{\Delta V}{I}$$

$$R = \rho \frac{l}{A}$$

Series:

$$V_t = V_1 + V_2 + \dots + V_n$$

$$R_t = R_1 + R_2 + \dots + R_n$$

Table of prefixes

Prefix	Symbol	Value
tera	T	10^{12}
giga	G	10^9
mega	M	10^6
kilo	k	10^3
centi	c	10^{-2}
milli	m	10^{-3}
micro	μ	10^{-6}
nano	n	10^{-9}
pico	p	10^{-12}